

Car Market Trends

Car Dekho Used-Vehicle Analysis — Full Project Documentation

Python · MySQL · Power BI Desktop

211 Cars · ₹6.24L Avg Price · 36.75% Avg Depreciation · India

TOOLS	SOURCE	RECORDS	SCOPE
Python · MySQL · Power BI	Car Dekho Used-Vehicle Listings	211 Cars + 88 Bikes	India, Prices in Lakhs INR

01 EXECUTIVE SUMMARY

- **Cleaned and analyzed 301 raw used-vehicle listings**, uncovering a critical scope issue at the source: 88 of 301 rows (29.2%) were actually motorbikes mislabeled as cars in a "car market" dataset — flagged, separated into a vehicle_type field, and excluded from the core 211-row car analysis rather than silently treated as cars.
- **Answered 9 structured pricing questions** covering depreciation, mileage, age, fuel type, transmission, seller type, and a direct bikes-vs-cars comparison — then reproduced every answer independently in MySQL across two related tables.
- **Built a 3-page, 15-visual interactive Power BI dashboard** with 4 DAX measures, a scatter-plot depreciation model, and a dedicated bikes-vs-cars comparison page — connected directly to MySQL.
- **Surfaced a specific, actionable pricing insight:** original list price (Present_Price) is a strong predictor of resale value ($r = 0.835$), but odometer reading is nearly irrelevant ($r = -0.059$) — while vehicle age ($r = -0.312$) and transmission type (Automatic sells for more than double Manual, ₹11.69L vs ₹5.33L) are the real drivers of resale price in this market.

02 PROBLEM STATEMENT

The Car Dekho dataset was distributed as a "car market" export of 301 used-vehicle listings with price, mileage, fuel type, and ownership fields — but a closer inspection revealed it silently mixed in a significant share of two-wheelers under names like "Royal Enfield Thunder 500" and "Honda Activa 4G." Treating these as cars would have meaningfully distorted every price and depreciation statistic, since two-wheelers sell in a fundamentally different price band. The goal of this project was to correctly scope the dataset to its stated subject (cars), quantify what drives used-car resale value, and only then use the excluded two-wheeler segment as a deliberate point of comparison rather than contamination.

03 OBJECTIVES

- **Obj 1:** Detect and correctly separate the mislabeled two-wheeler listings before any price analysis, rather than after.
- **Obj 2:** Quantify the relationship between original price, resale price, mileage, and age using correlation analysis.

- **Obj 3:** Compare pricing across fuel type, transmission, and seller type to identify the strongest categorical price drivers.
- **Obj 4:** Reproduce all findings independently in MySQL across both the cars-only and full (cars+bikes) datasets.
- **Obj 5:** Build a Power BI dashboard that treats the bikes-vs-cars comparison as its own dedicated view rather than blending it into the main pricing pages.
- **Obj 6:** Produce a portfolio-ready deliverable demonstrating rigorous dataset-scoping judgment, not just mechanical cleaning steps.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Car Dekho Used-Vehicle Listings
File Format	CSV, single table
Raw Records	301 rows × 9 columns
Two-Wheelers Identified	88 rows (29.2%) — flagged via vehicle_type, excluded from core analysis
Cars-Only Records	211 rows (after also removing 2 exact duplicates)
Currency & Units	Prices in lakhs INR (1 lakh = 100,000 rupees)
Missing Values	Zero across all 9 columns
Notable Outlier (resolved)	A 500,000 km odometer reading traced to "Activa 3g" — correctly a bike, not a data error

Key Columns Used in Analysis

Column	Type	Description
car_name	Text	Model name (98 unique values across the full dataset)
vehicle_type	Text	Derived: 'car' or 'bike', based on name-keyword matching
selling_price	Decimal	Resale price in lakhs INR (\$ ~1.2L avg for bikes, ~6.2L for cars)
present_price	Decimal	Original list price in lakhs INR
kms_driven	Integer	Odometer reading (cars-only max: 197,176 km)
age	Integer	Derived: 2018 – Year (2018 = latest year in the data)
price_drop_pct	Decimal	Derived: (present_price – selling_price) / present_price × 100
fuel_type / transmission / seller_type	Text	Petrol/Diesel/CNG · Manual/Automatic · Dealer/Individual

05 TOOLS & TECHNOLOGIES USED

Tool	Platform	Purpose in This Project
Python 3	VS Code	Bike/car detection, cleaning, correlation analysis, visualization

Tool	Platform	Purpose in This Project
pandas / numpy	Python library	Keyword-based vehicle_type flagging, derived-column engineering
MySQL 8	MySQL Workbench	Two related tables (cars, vehicles_all); SQL validation of all findings
Power BI Desktop	Microsoft	3-page dashboard including a dedicated bikes-vs-cars comparison page
DAX	Power BI native language	4 measures: Total Cars, Avg Selling Price, Avg Depreciation %, Avg Kms Driven

06 DATA CLEANING PROCESS — PYTHON

The most consequential cleaning decision in this project happens before any numeric cleaning — correctly scoping which rows are actually cars.

Step	Action	Result
1	Load CSV, confirm zero missing values, check for duplicates	301 rows × 9 columns, 2 duplicates found
2	Keyword-match car_name against known bike brands/models, flag vehicle_type	88 of 301 rows (29.2%) identified as bikes
3	Drop 2 exact duplicate rows	299 rows remain (across both vehicle types)
4	Filter to vehicle_type='car' for the core analysis	211 cars-only rows
5	Derive age = 2018 – Year	Interpretable vehicle age instead of raw model year
6	Derive price_drop_pct = (present_price – selling_price) / present_price × 100	Direct depreciation metric, avg 36.75%

07 SQL / MYSQL — VALIDATION LAYER

Two related tables: cars (211 rows, the core pricing analysis) and vehicles_all (299 rows, cars + bikes, used only for the Q9 comparison). Both were checked in Model view to confirm no auto-detected relationship was created between them, since they answer structurally different questions despite sharing column names.

Reserved-Word-Free Schema, One Real SQL Gotcha

Unlike other projects in this series, Car Dekho's SQL layer needed no Pearson-formula workaround for its 3 correlation questions (present vs. selling price, kms vs. price, age vs. price) — all handled with the same reusable formula pattern used elsewhere. The more interesting finding was a genuine tie: 'amaze' and 'jazz' are both listed exactly 7 times, tying for the 10th spot in the Top 10 Most Listed Models. Python and MySQL, run without an explicit tiebreaker, legitimately returned different (both correct) 10th-place models — resolved by adding a secondary ORDER BY car_name ASC for deterministic output.

Analytical VIEW

A vw_cars_summary view buckets cars by age (4 bands) and price (4 bands) crossed with fuel_type, transmission, and seller_type — enabling a VIEW Weighted Avg Selling Price measure that cross-validates the row-level average.

08 DAX MEASURES CREATED

Measure	DAX Formula	Result
Total Cars	COUNTROWS(cars)	211
Avg Selling Price	AVERAGE(cars[selling_price])	₹6.24 lakh
Avg Depreciation %	AVERAGE(cars[price_drop_pct])	36.75
Avg Kms Driven	AVERAGE(cars[kms_driven])	40,223 km
VIEW Weighted Avg Selling Price	SUMX-weighted average from vw_cars_summary	₹6.24 lakh (cross-check)

Formatting note: Avg Depreciation % is already a percentage-scale number coming out of Python (not a 0–1 ratio), so it must be formatted as Custom (0.00"%) rather than Power BI's Percentage type — which would multiply an already-scaled 36.75 by 100 again.

09 DASHBOARD VISUALS BUILT (15 Visuals, 3 Pages)

Page 1 — Pricing Overview

#	Visual Type	Fields Used	Insight Delivered
1-4	KPI Cards	Total Cars / Avg Selling Price / Avg Depreciation % / Avg Kms Driven	211 cars, ₹6.24L avg, 36.75% avg depreciation, 40,223 km avg
5	Scatter + trend line	X: present_price / Y: selling_price (Don't summarize)	Strong positive relationship, corr = 0.835
6	Bar Chart	X: fuel_type / Y: Avg Selling Price	Diesel commands the highest resale price
7	Bar Chart	X: transmission / Y: Avg Selling Price	Automatic more than doubles Manual pricing
8-10	Slicers	fuel_type, transmission, seller_type	Cross-filters all Page 1 visuals

Page 2 — Depreciation & Listings

#	Visual Type	Fields Used	Insight Delivered
1	Histogram (binned column)	price_drop_pct distribution	Depreciation peaks in the 25–35% range
2	Scatter	X: kms_driven / Y: selling_price	Visually confirms the near-zero correlation
3	Scatter	X: age / Y: selling_price	Visible downward trend, corr = -0.312
4	Bar Chart (Top 10 filtered)	Y: car_name / X: Total Cars	'city' leads at 26 listings
5	Slicer	seller_type	Cross-filters Page 2 visuals

Page 3 — Bikes vs. Cars (from vehicles_all)

- Bar Chart comparing Avg Selling Price and Avg Kms Driven by vehicle_type — cars average ₹6.24L vs. bikes' ₹0.64L, a ~10x price gap, while cars are also driven further on average (40,223 km vs. 28,989 km).

10 KEY INSIGHTS & RESULTS

A. Dataset Scope

- **Nearly 3 in 10 listings in a 'car' dataset were actually motorbikes:** 88 of 301 rows (29.2%) — a scoping issue significant enough to invalidate any price statistic computed before it was caught.

B. What Actually Predicts Resale Price

- **Original price is the strongest predictor of resale value:** $r = 0.835$ between `present_price` and `selling_price` — unsurprising, but a critical sanity-check baseline.
- **Odometer reading is nearly irrelevant:** $r = -0.059$ between `kms_driven` and `selling_price` — a genuinely surprising, counter-intuitive finding worth flagging rather than assuming mileage must matter.
- **Vehicle age matters moderately:** $r = -0.312$ — a real but not dominant effect, weaker than transmission type's impact.

C. Categorical Price Drivers

- **Transmission type shows the largest categorical gap:** Automatic averages ₹11.69 lakh versus Manual's ₹5.33 lakh — more than double.
- **Fuel type and seller type both show large, though possibly confounded, gaps:** Diesel (₹10.10L) vastly outsells Petrol (₹4.79L) and CNG (₹3.10L); Dealers (₹6.63L) far outprice Individuals (₹2.00L) — both likely reflect a correlation with vehicle segment/size rather than fuel or seller type directly causing the price difference.

D. Bikes vs. Cars

- **Cars sell for roughly 10x what bikes do in this market:** ₹6.24L average for cars versus ₹0.64L for bikes — confirming the original scoping decision to separate them was essential, not optional.

11 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — A Mislabeled Dataset Scope

Problem: 29.2% of rows in a 'car market' dataset were actually two-wheelers, discoverable only by inspecting the actual `car_name` values rather than trusting the dataset's stated subject.

Solution: Built a keyword-matching filter against known bike brands/models, created an explicit `vehicle_type` field, and made the exclusion decision visible and documented rather than silent — then used the excluded segment productively as its own comparison rather than discarding it.

Challenge 2 — An Outlier That Turned Out Not to Be One

Problem: A 500,000 km odometer reading looked like a likely data-entry error worth investigating or capping.

Solution: Traced the specific row before assuming it was bad data — it belonged to an "Activa 3g," correctly excluded as a bike, closing the question rather than applying an arbitrary cap that might have hidden a real, legitimate high-mileage car elsewhere.

Challenge 3 — A Genuine Tie at the Reporting Boundary

Problem: Python and MySQL returned different 10th-place models ('amaze' vs. 'jazz') for the Top 10 Most Listed Models question, both tied at exactly 7 listings.

Solution: Verified both were correct rather than assuming one tool had a bug, then added a secondary `ORDER BY car_name ASC` in SQL for reproducible, deterministic output going forward.

Challenge 4 — A Depreciation Measure Missing Its Percent Sign

Problem: Avg Depreciation % initially rendered as a bare "36.75" in Power BI, with no visual indication it was a percentage.

Solution: Applied a Custom number format (0.00"%") rather than the built-in Percentage type, since the underlying value was already percentage-scaled coming out of Python.

12 SKILLS LEARNED & TAKEAWAYS

- Dataset-scope verification as a first-class cleaning step: checking whether rows actually match the dataset's stated subject before analyzing them, not just checking for nulls and duplicates.
- Resisting the urge to cap or remove outliers before tracing their source — the 500,000 km value was correctly excluded for an entirely different reason than mileage error.
- Treating an excluded data segment as a deliberate comparison group (bikes vs. cars) rather than simply discarding it once it served its exclusion purpose.
- Correlation-strength judgment: distinguishing which relationships are strong and expected (present vs. selling price) from which are surprisingly weak (mileage) and worth calling out explicitly.
- Handling genuine ties at a Top-N boundary with a deterministic secondary sort, rather than treating tool disagreement as a bug to chase.
- The specific DAX formatting distinction between a ratio needing *100 and an already-scaled percentage needing only a display symbol.

13 CONCLUSION

- **Correctly scoped 301 raw listings down to 211 genuine cars**, catching and separating a 29.2% two-wheeler contamination rate before any price statistic was computed — the single most consequential decision in the entire project.
- **Independently validated all findings in MySQL across two related tables**, including confirming a genuine tie at the Top-10 boundary rather than assuming a tool discrepancy was an error.
- **Delivered a 3-page, 15-visual Power BI dashboard**, with a dedicated page giving the excluded bikes segment its own legitimate comparison rather than leaving it unused.
- **Most notably, let a counter-intuitive correlation stand as reported:** odometer reading ($r = -0.059$) turned out to barely matter for resale price in this market, while transmission type and vehicle age mattered considerably more — a finding only visible because the dataset was correctly scoped first.

KPI Summary

Metric	Value	Metric	Value
Total Cars (analyzed)	211	Bikes Excluded	88 (29.2%)
Avg Selling Price	₹6.24 lakh	Avg Depreciation	36.75%
Present ↔ Selling Price Corr.	0.835	Kms Driven ↔ Price Corr.	-0.059
Age ↔ Price Corr.	-0.312	Avg Kms Driven	40,223 km
Top Fuel Type (Price)	Diesel — ₹10.10L	Top Transmission (Price)	Automatic — ₹11.69L

Metric	Value	Metric	Value
Most-Listed Model	city — 26 listings	Cars vs Bikes Price Gap	~10x (₹6.24L vs ₹0.64L)
Duplicates Removed	2	Missing Values	0